REFLOWECONOMY · USER REQUIREMENTS

ReflowEconomy user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First user requirements (kit 1.8 pipeline), drawn from the users-and-needs and stakeholder tables of RFE-PRB-001 v0.6

ReflowEconomy user requirements

What the people who would run, work in, sell to and regulate a micro-factory need, written from their side. Product requirements that meet these needs are in RFE-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix RFE-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22. These are first-pass needs taken from the problem statement and its sources; none has yet been confirmed in a co-design session with the first partner (a waste-picker cooperative; first candidate SWaCH, Pune, nothing agreed), and all are to be revised from what that partner says.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	What to process here	We need to know which materials we can recover safely ourselves and which must go to a licensed refiner	High	Micro-factory operator; municipality (PRB Tables 1 and 2)
UR-002	Usability	Recipes we can follow	We need step-by-step recipes for each material we process: what goes in, temperatures, what comes out, what protection to wear	High	Micro-factory operator; line worker (PRB Table 1)
UR-003	Cost	Know if it pays	Before we spend money, we need to see whether a site can pay its way at our own prices, wages and tariff	High	Micro-factory operator (PRB Table 1)
UR-004	Logistics	Send away safely	We need to know how to pack, label and where to send what we cannot process	Medium	Industrial refiner or exporter (PRB Table 1)
UR-005	Deployment and assembly	Fit a small site	It has to fit in a rented shed or two shipping containers of about 60 m², with room to move safely	High	Micro-factory operator (PRB Table 1)
UR-006	Performance	Handle a day's collection	It must handle what a cooperative brings in a day, about 100 kg a shift	High	Micro-factory operator; waste picker (PRB Table 1)
UR-007	Performance	Keep value local	Most of the value of what we collect should stay with us and our community, not leave as cheap scrap	High	Waste picker or cooperative member; municipality (PRB Tables 1 and 2)
UR-008	Environment and durability	Little to disposal, nothing burned	As little as possible should go to disposal, and nothing is ever burned	High	Municipality; waste picker (PRB Tables 1 and 2)
UR-009	Cost	Low electricity bill	The electricity bill must not eat the margin	Medium	Micro-factory operator (PRB Table 1)
UR-010	Deployment and assembly	Use the supply we can get	It must run on the electricity connection a small site can actually get	High	Micro-factory operator; electricity distribution company (PRB Tables 1 and 2)
UR-011	Safety	Clean air	We must be able to work a full shift without breathing harmful fumes	High	Line worker (PRB Table 1)
UR-012	Safety	Safe machines	We must be able to work near the shredder, hot presses, noise and electricity without being hurt	High	Line worker (PRB Table 1)
UR-013	Regulatory	Trusted material	Buyers need to trust what they buy: polymer, grade, contamination and where it came from	High	Local buyer of recycled material (PRB Table 1)
UR-014	Usability	Quick records	Recording a lot must not slow the work	Medium	

ID	Category	Title	Need	Priority	Source
					Micro-factory operator (PRB Table 1)
UR-015	Maintenance	Build and repair locally	The machines and the fit-out must be things a local fabricator can make and repair	Medium	Micro-factory operator; open hardware community (PRB Table 1)
UR-016	Cost	Affordable equipment	The equipment must be within reach of a cooperative's capital	Medium	Micro-factory operator (PRB Table 1)
UR-017	Regulatory	Permittable	The authorities must be able to register and permit the site	High	Municipality; state pollution control board (PRB Table 2)
UR-018	Safety	Fire safety	We must be protected from fires, including fires started by lithium cells hidden in the input	High	Line worker; municipality (PRB Tables 1 and 2)
UR-019	Usability	Fair intake rules	Rules for refusing or charging for poor loads must be clear and fair to pickers	Medium	Waste picker or cooperative member (PRB Table 1)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.